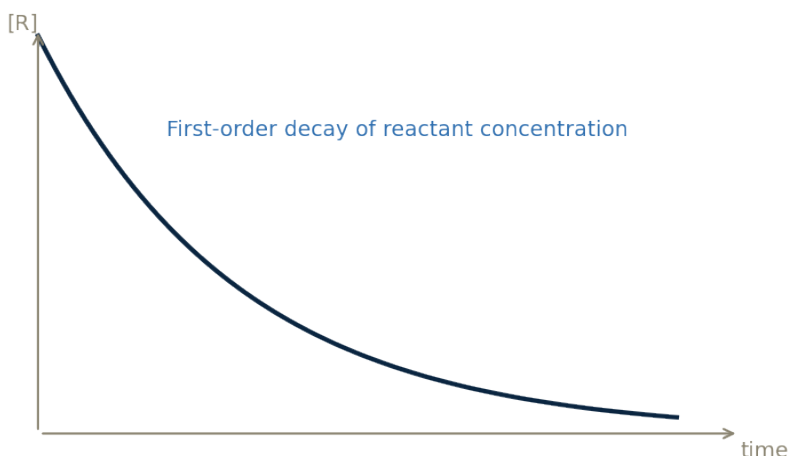


Chemical Kinetics

JEE Chemistry — Notes, Formula Sheet, Practice Problems & Diagram

1. Notes

- Rate of reaction = change in concentration of reactant/product per unit time. Rate = $-(1/a)(d[A]/dt) = +(1/c)(d[C]/dt)$ for $aA \rightarrow cC$.
- Rate law: rate = $k[A]^m[B]^n$, where m and n are the orders with respect to A and B (determined experimentally, not from the balanced equation).
- For a zero-order reaction, rate is independent of concentration; for a first-order reaction, rate is directly proportional to concentration of one reactant.
- Integrated rate law for first order: $\ln[A] = \ln[A]_0 - kt$, or $[A] = [A]_0 e^{-kt}$. Half-life for first order is constant: $t(1/2) = 0.693/k$.
- Arrhenius equation: $k = A e^{-E_a/RT}$, showing that rate constant increases with temperature and decreases with higher activation energy (E_a).
- Collision theory: reactions occur when molecules collide with sufficient energy ($\geq$ activation energy) and correct orientation.
- A catalyst increases reaction rate by providing an alternate pathway with lower activation energy, without being consumed in the reaction.



First-order reaction: concentration of reactant decays exponentially with time.

2. Formula Sheet

Rate law	$\text{Rate} = k [\text{A}]^m [\text{B}]^n$
Zero-order integrated rate law	$[\text{A}] = [\text{A}]_0 - kt$
First-order integrated rate law	$\ln[\text{A}] = \ln[\text{A}]_0 - kt$
Half-life (first order)	$t(1/2) = 0.693 / k$
Half-life (zero order)	$t(1/2) = [\text{A}]_0 / (2k)$
Arrhenius equation	$k = A e^{(-E_a/RT)}$
Arrhenius equation (log form)	$\ln k = \ln A - E_a/(RT)$
Order of reaction	Overall order = $m + n$ (sum of individual orders)

3. Practice Problems (22 questions)

1. A first-order reaction has a rate constant $k = 0.0693 \text{ /min}$. Find its half-life.
2. For a reaction, $\text{rate} = k[\text{A}]^2[\text{B}]$. Find the overall order of the reaction.
3. A first-order reaction has initial concentration 0.4 M . Find the concentration remaining after 2 half-lives.
4. State the difference between the rate law and the balanced chemical equation, and why the rate law must be determined experimentally.
5. A zero-order reaction has rate constant $k = 0.05 \text{ mol/(L min)}$ and initial concentration 1 M . Find the time for the concentration to become zero.
6. Explain the effect of a catalyst on the activation energy and rate of a reaction.
7. Find the rate constant of a first-order reaction whose half-life is 20 minutes.
8. State collision theory and explain why not all collisions between reactant molecules lead to a reaction.
9. For a first-order reaction, if the initial concentration is doubled, how does the half-life change?
10. Find the activation energy of a reaction if its rate constant doubles when temperature increases from 300 K to 310 K (use the Arrhenius equation qualitatively/approximate reasoning).
11. Explain the difference between the order of a reaction and its molecularity.
12. A reaction is first order with respect to A and second order with respect to B. Write its rate law.
13. Find the time required for a first-order reaction with $k = 0.023 \text{ /s}$ to reduce concentration from 0.8 M to 0.2 M (using $\ln[\text{A}] = \ln[\text{A}]_0 - kt$, $\ln 4$ approx 1.386).
14. State why the rate of most reactions increases with an increase in temperature, using collision theory.
15. For a zero-order reaction, sketch (describe) the shape of the concentration vs time graph.
16. Explain the significance of activation energy (E_a) in the Arrhenius equation.
17. A first-order reaction is 50% complete in 20 minutes. Find the rate constant k .

18. Explain why increasing the concentration of reactants generally increases the rate of a reaction (in terms of collision frequency).
19. State the units of the rate constant k for a zero-order and a first-order reaction.
20. Explain why a reaction with a very high activation energy proceeds slowly at room temperature.
21. For a second-order reaction, write the general rate law in terms of a single reactant A.
22. Explain the difference between an elementary reaction and a complex (multi-step) reaction, in terms of molecularity and order.

Answer Key

1. $t(1/2) = 10 \text{ min}$
2. Order = 3 (third order overall)
3. Concentration = 0.1 M (after 2 half-lives: $0.4 \rightarrow 0.2 \rightarrow 0.1$)
4. Conceptual - rate law depends on the mechanism, not stoichiometry; must be found experimentally
5. $t = 20 \text{ min}$ (using $t = [A]_0/k = 1/0.05$)
6. Conceptual - catalyst lowers E_a by providing an alternate pathway, increasing rate without being consumed
7. $k = 0.0347 \text{ /min}$ (approx, using $k = 0.693/t(1/2)$)
8. Conceptual - collisions need sufficient energy ($\geq E_a$) and correct orientation; most collisions fail one or both conditions
9. Conceptual - half-life for first order is independent of initial concentration (constant)
10. Conceptual - increasing T increases k (rate constant) per Arrhenius equation; specific E_a calculation requires the full equation with both T values
11. Conceptual - order is empirical (from rate law); molecularity is the number of species colliding in an elementary step (theoretical)
12. Rate = $k[A][B]^2$
13. $t \text{ approx } 60.3 \text{ s}$ (using $t = \ln 4/k = 1.386/0.023$)
14. Conceptual - higher T increases average kinetic energy, so more molecules have energy $\geq E_a$, increasing collision frequency with sufficient energy
15. Conceptual - straight line with negative slope (linear decrease) for zero order
16. Conceptual - E_a is the minimum energy barrier that colliding molecules must overcome for a reaction to occur
17. $k = 0.0347 \text{ /min}$
18. Conceptual - higher concentration means more molecules per unit volume, increasing collision frequency and hence rate
19. Conceptual - zero order: $\text{mol L}^{-1} \text{ s}^{-1}$; first order: s^{-1}
20. Conceptual - very few molecules have enough energy to overcome a high E_a at room temperature, so reaction rate is very low
21. Rate = $k[A]^2$
22. Conceptual - elementary reaction occurs in a single step, with molecularity = order; complex reactions occur via multiple steps, with order determined by the slowest (rate-determining) step

Note: conceptual/derivation questions are marked accordingly in the answer key. Numerical answers are rounded for clarity — show full working in your own solutions.